

6 Conformal Prediction

So far, our uncertainty estimates have been probabilities for binary outcomes. In many applications the label space is much larger: a real-valued response, one of many classes, or even a structured object. A natural uncertainty estimate in these settings is a *prediction set* containing the labels that remain plausible.

Let $\mathcal{Y}$ be an arbitrary label space. A prediction-set function is a map $T: \mathcal{X} \rightarrow 2^{\mathcal{Y}}$. For regression, $T(x)$ is often an interval. For multiclass classification, it is a subset of the possible classes. We would like the set to contain the true label with probability at least $1 - \delta$:

$$\mathbb{P}_{(X,Y) \sim \mathcal{D}} [Y \in T(X)] \geq 1 - \delta.$$

This is a *marginal coverage* guarantee: the probability averages over both X and Y . Of course, the uninformative choice $T(x) = \mathcal{Y}$ always has coverage one. The goal is to obtain coverage while keeping the sets small and useful.

6.1 Nonconformity scores

Conformal prediction reduces the problem of constructing a set in $\mathcal{Y}$ to the one-dimensional problem of choosing a threshold. It starts with a *nonconformity score*

$$s: \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R},$$

where a larger $s(x, y)$ means that the candidate label y is less compatible with the features x . Every threshold τ defines a prediction set

$$T_{s,\tau}(x) := \{y \in \mathcal{Y} : s(x, y) \leq \tau\}.$$

Example 6.1 (Regression residuals). Given a point predictor $h: \mathcal{X} \rightarrow \mathbb{R}$, take

$$s(x, y) = |y - h(x)|.$$

Then

$$T_{s,\tau}(x) = [h(x) - \tau, h(x) + \tau].$$

The conformal procedure will choose the radius τ from held-out data. This simple score produces intervals of the same width for every x .

Example 6.2 (Adapting to heterogeneous noise). Suppose a model also estimates a positive scale $\sigma(x)$ describing how noisy the response is near x . The normalized score

$$s(x, y) = \frac{|y - h(x)|}{\sigma(x)}$$

produces

$$T_{s,\tau}(x) = [h(x) - \tau\sigma(x), h(x) + \tau\sigma(x)].$$

The threshold is still one-dimensional, but the interval is wider on examples the underlying model

considers difficult. Conformal prediction calibrates the score; the choice of score determines how informative the resulting sets can be.

Example 6.3 (Multiclass classification). Suppose a classifier assigns a score $\widehat{p}_y(x) \in [0, 1]$ to every class y . One possible nonconformity score is

$$s(x, y) = 1 - \widehat{p}_y(x).$$

The resulting set contains every label whose model score exceeds $1 - \tau$. More refined scores can take account of the cumulative model score of labels ranked ahead of y , but the coverage theorem below does not require the scores to be accurate probabilities.

6.2 Split conformal prediction

Suppose the score s has already been fixed, perhaps by fitting a model on a training set. We now receive an independent *calibration set*

$$D_{\text{cal}} = ((X_1, Y_1), \dots, (X_n, Y_n))$$

and compute the calibration scores $S_i = s(X_i, Y_i)$. Let

$$k := \lceil (n + 1)(1 - \delta) \rceil.$$

Algorithm 4: Split conformal prediction

Input: A fixed nonconformity score s , calibration data D_{cal} of size n , and a target miscoverage rate $\delta \in (0, 1)$.

1. Compute $S_i = s(X_i, Y_i)$ for $i = 1, \dots, n$ and set $k = \lceil (n + 1)(1 - \delta) \rceil$.
2. If $k \leq n$, let $\widehat{\tau}$ be the k th smallest value among $S_1, \dots, S_n$. If $k = n + 1$, set $\widehat{\tau} = +\infty$.

Output: The prediction-set function

$$T(x) = \{y \in \mathcal{Y} : s(x, y) \leq \widehat{\tau}\}.$$

The use of $n + 1$, rather than n , is the finite-sample correction that makes the guarantee exact. The extra point is the future test example itself.

Theorem 6.4 (Marginal coverage of split conformal). *Suppose the n calibration examples and a test example (X_{n+1}, Y_{n+1}) are i.i.d., and the score s is fixed independently of these examples. If the $n + 1$ scores are almost surely distinct, the output of Algorithm 4 satisfies*

$$1 - \delta \leq \mathbb{P}[Y_{n+1} \in T(X_{n+1})] \leq 1 - \delta + \frac{1}{n + 1}.$$

Proof. Let $S_{n+1} = s(X_{n+1}, Y_{n+1})$. Exchangeability and the absence of ties imply that the rank R of S_{n+1} among $S_1, \dots, S_{n+1}$ is uniform on $\{1, \dots, n+1\}$.

When $k \leq n$, the test label is covered exactly when

$$S_{n+1} \leq \widehat{\tau},$$

which occurs exactly when $R \leq k$. Therefore

$$\mathbb{P}[Y_{n+1} \in T(X_{n+1})] = \mathbb{P}[R \leq k] = \frac{k}{n+1}.$$

Since $k = \lceil (n+1)(1-\delta) \rceil$,

$$1-\delta \leq \frac{k}{n+1} < 1-\delta + \frac{1}{n+1}.$$

When $k = n+1$, the threshold is infinite and coverage is one, which obeys the same weak inequalities. $\square$

Independent and identically distributed examples are exchangeable, but exchangeability is the weaker property actually used in the proof: permuting the $n+1$ examples must not change their joint distribution. No assumption is made about whether the model is correct, whether s has a known distribution, or how complicated the feature space is.

Example 6.5 (The finite-sample correction). Suppose $n = 19$ and we want 90% coverage, so $\delta = 0.1$. Then

$$k = \lceil 20 \cdot 0.9 \rceil = 18.$$

We use the 18th smallest of the 19 calibration scores. The test score has one of 20 equally likely ranks and is covered for ranks 1 through 18, giving coverage exactly $18/20 = 0.9$ when there are no ties.

6.3 Group conditional conformal prediction

Marginal coverage can hide large differences between different parts of the population. We might instead want a prediction set satisfying

$$\mathbb{P}[Y \in T(X) \mid g(X) = 1] \geq 1 - \delta$$

for a group $g: \mathcal{X} \rightarrow \{0, 1\}$. We can obtain this guarantee directly by applying split conformal prediction only to the calibration examples belonging to the group.

Fix a group g independently of the calibration data, and let

$$I_g := \{i \in \{1, \dots, n\} : g(X_i) = 1\}, \quad n_g := |I_g|,$$

and

$$k_g := \lceil (n_g + 1)(1 - \delta) \rceil.$$

If $k_g \leq n_g$, let $\widehat{\tau}_g$ be the k_g th smallest score among $\{S_i : i \in I_g\}$. If $k_g = n_g + 1$, set $\widehat{\tau}_g = +\infty$. The group-specific prediction set is

$$T_g(x) := \{y \in \mathcal{Y} : s(x, y) \leq \widehat{\tau}_g\}.$$

Theorem 6.6 (Coverage within a fixed group). *Suppose the calibration examples and the test example are independent and identically distributed, and the score s and group g are fixed independently of them. If $\mathbb{P}[g(X) = 1] > 0$, then*

$$\mathbb{P}[Y_{n+1} \in T_g(X_{n+1}) \mid g(X_{n+1}) = 1] \geq 1 - \delta.$$

If the scores have no ties, then conditional on $n_g = m$, the coverage is $k_g/(m + 1)$ when $k_g \leq m$, and is one when $k_g = m + 1$.

Proof. Condition on which calibration examples belong to g , on $n_g = m$, and on the event $g(X_{n+1}) = 1$. The m selected calibration examples and the test example are independent draws from the distribution of (X, Y) conditional on $g(X) = 1$. Their scores are therefore exchangeable. The same rank argument as in Theorem 6.4, now with m calibration scores, gives the result. $\square$

The simplest application is when $g_1, \dots, g_M$ form a partition of the feature space: every x belongs to exactly one group. We compute a separate threshold $\widehat{\tau}_j$ using the calibration examples in group g_j , and use

$$T(x) = \{y \in \mathcal{Y} : s(x, y) \leq \widehat{\tau}_j\} \quad \text{when } g_j(x) = 1.$$

This procedure, often called *Mondrian conformal prediction*, guarantees coverage at least $1 - \delta$ within every fixed group g_j . For example, one could separately calibrate prediction intervals for several regions, age ranges, or classes.

We can also handle a fixed finite collection $\mathcal{G}$ of possibly overlapping groups. Calibrate a threshold $\widehat{\tau}_g$ separately for every $g \in \mathcal{G}$ and, assuming the groups cover $\mathcal{X}$, define

$$\widehat{\tau}(x) := \max_{g \in \mathcal{G}: g(x)=1} \widehat{\tau}_g, \quad T(x) := \{y \in \mathcal{Y} : s(x, y) \leq \widehat{\tau}(x)\}.$$

For any fixed $g \in \mathcal{G}$, if $g(x) = 1$, then $\widehat{\tau}(x) \geq \widehat{\tau}_g$, so $T(x) \supseteq T_g(x)$. Consequently, Theorem 6.6 immediately gives

$$\mathbb{P}[Y \in T(X) \mid g(X) = 1] \geq 1 - \delta \quad \text{for every fixed } g \in \mathcal{G}.$$

There is no union bound in this argument. Each displayed inequality averages over both the random calibration set and the random test example; it does not say that, with high probability over one realized calibration set, all of the corresponding conditional coverage probabilities are simultaneously large.

This direct construction has two important costs. First, the maximum can make the sets quite conservative when many groups overlap. Second, a small group supplies little calibration data. Indeed, whenever $k_g = n_g + 1$, its threshold is infinite and every prediction set for a member of that group becomes uninformative. More sophisticated methods seek useful coverage guarantees for rich, overlapping group families without simply taking this maximum; see, for example, Bairaktari, Wu, and Wu [BWW25].

Acknowledgments and AI Use

The overall presentation is more or less AI-free, but I used AI extensively to write up the notes.